0-SM PP / CP / Engram Primitives

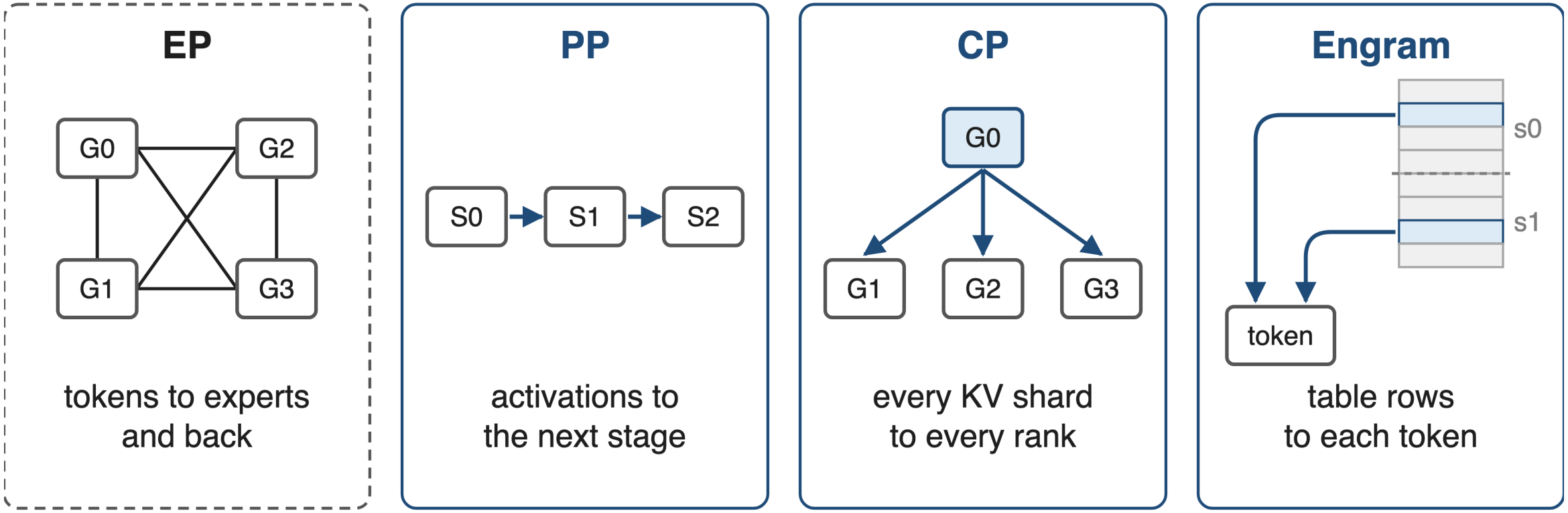
Let the NIC and copy engines move the bytes

DeepSeek-AI · DeepEP V2 (Apr 2026) and V2.5 (Sep 2026)

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Expert-Parallel Communication · github.com/deepseek-ai/DeepEP

DeepEP now moves bytes for PP, CP and Engram



Blue: experimental primitives added in DeepEP V2. Engram is DeepSeek's hashed N-gram memory (arXiv 2601.07372).

Communication holds 15% of the SMs

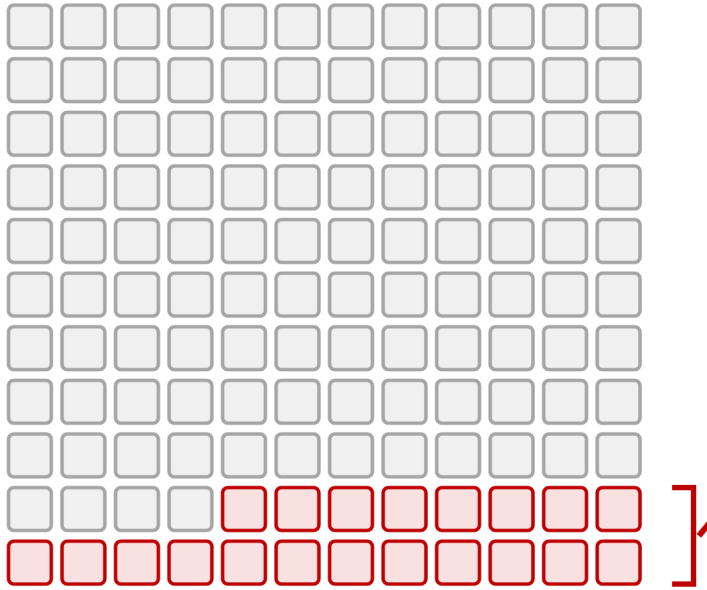
20 / 132

H800 SMs that DeepSeek-V3 reserves for communication; DeepEP's 20 SMs cost ~20% GEMM efficiency

DeepSeek-V3 report, arXiv 2412.19437, §3.5.1 (p.20); Megatron-Core MoE report, arXiv 2603.07685, §4.2.3 (p.34)

Communication SMs mostly copy and wait

one H800 GPU: 132 SMs



compute communication

what the communication SMs do

- copy between RDMA buffers and tensors
- forward IB traffic into NVLink
- reduce expert outputs in combine
- lay out tokens per expert
- poll flags until data lands

tensor cores on these SMs sit idle

Task list from the DeepSeek-V3 report §3.5.1; flag polling as in DeepEP's kernels

Fewer SMs is not zero SMs

- **Shrink the kernels** (DeepEP V2 EP)
- 24 → **4–6** SMs
- **Combine sums, dispatch routes**: SMs stay

- **Offload to hardware** (V3 wish list)
- Comm co-processor, SHARP-style
- **Not in today's GPUs**

Needed: traffic whose data path has no arithmetic

Insight: moving bytes needs no SM

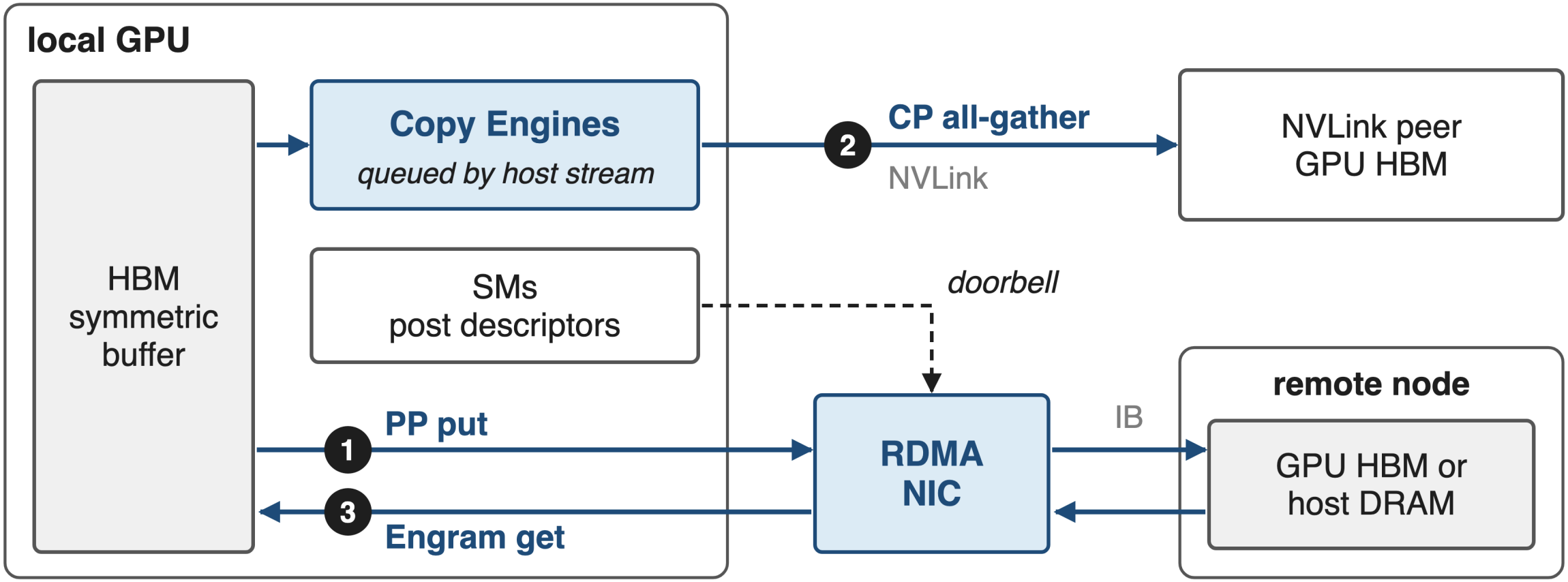
- PP sends, CP all-gather, Engram lookups only **copy**
- Addresses known before launch: no sum, no routing
- Nodes already have byte movers: **RDMA NIC**, **copy engines**

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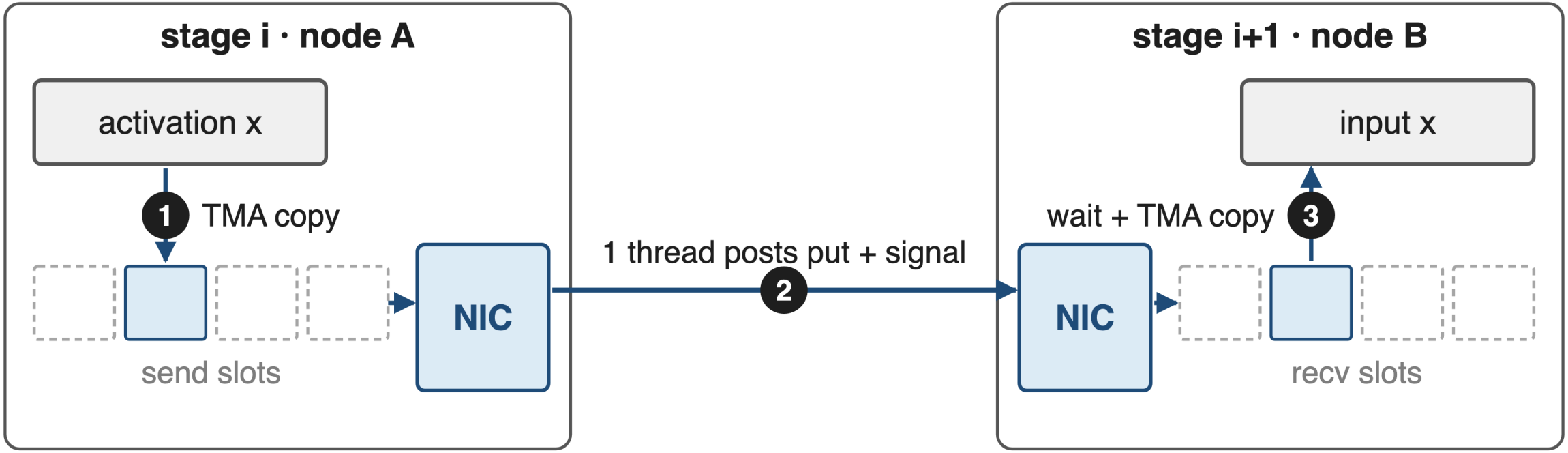
SMs only post descriptors; the NIC and copy engines move the payload

Each primitive rides a byte mover

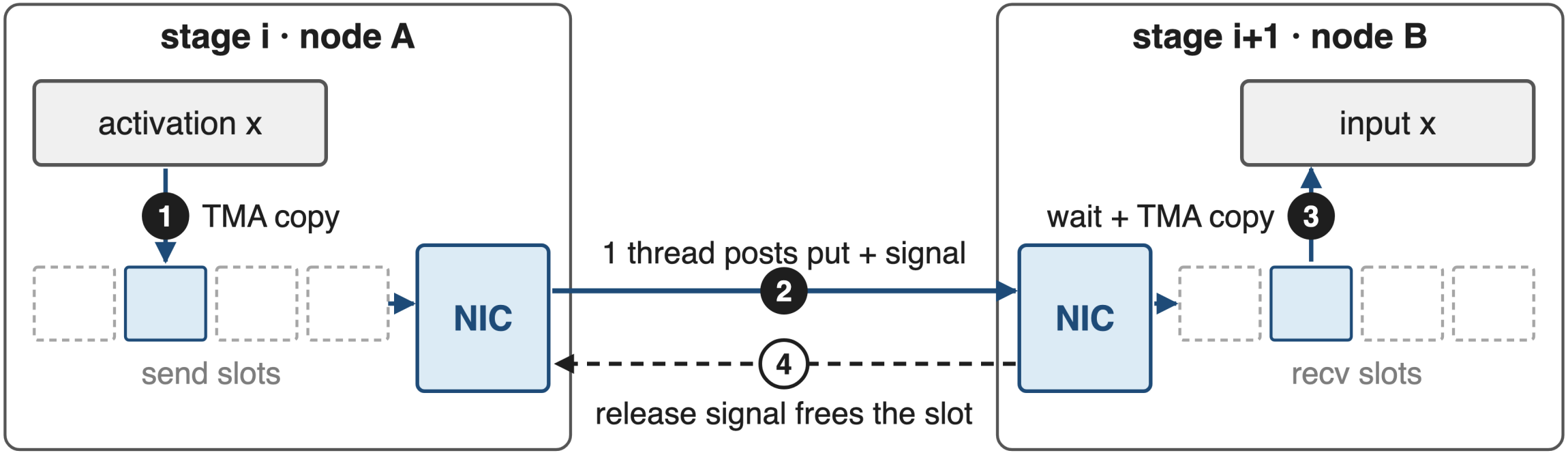


Doorbell = a GPU thread writes the work request and rings the NIC (NCCL Gin, GDAKI). Engram reads are one-sided: the row owner runs nothing.

❶ PP: a thread posts, the NIC sends

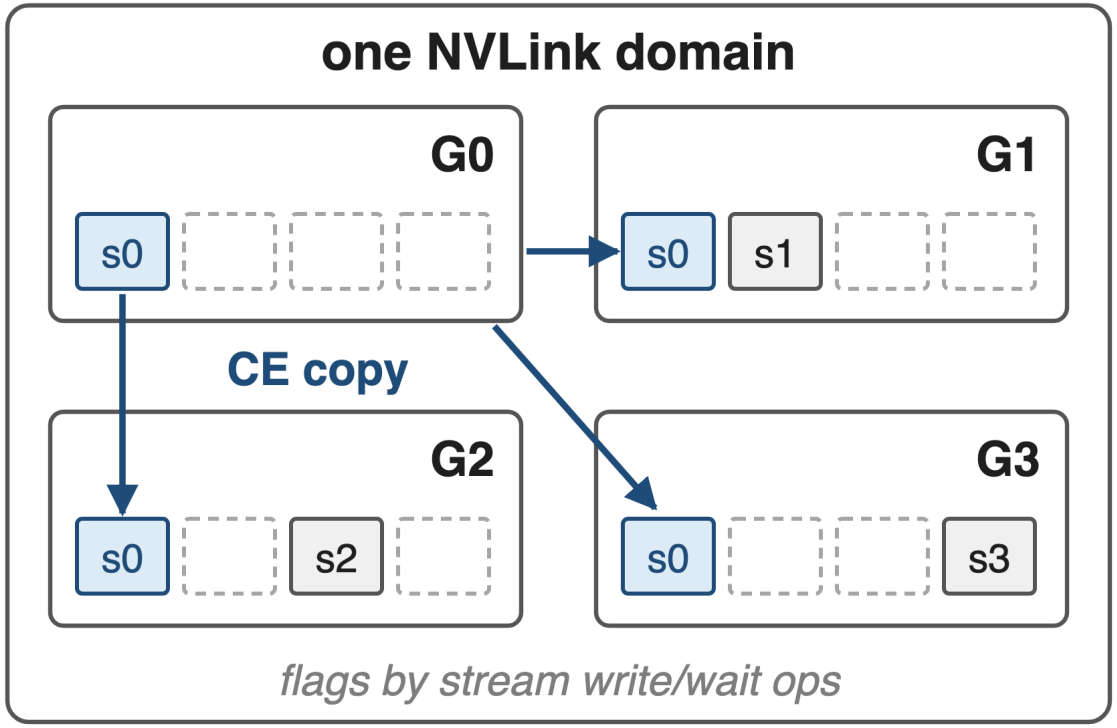


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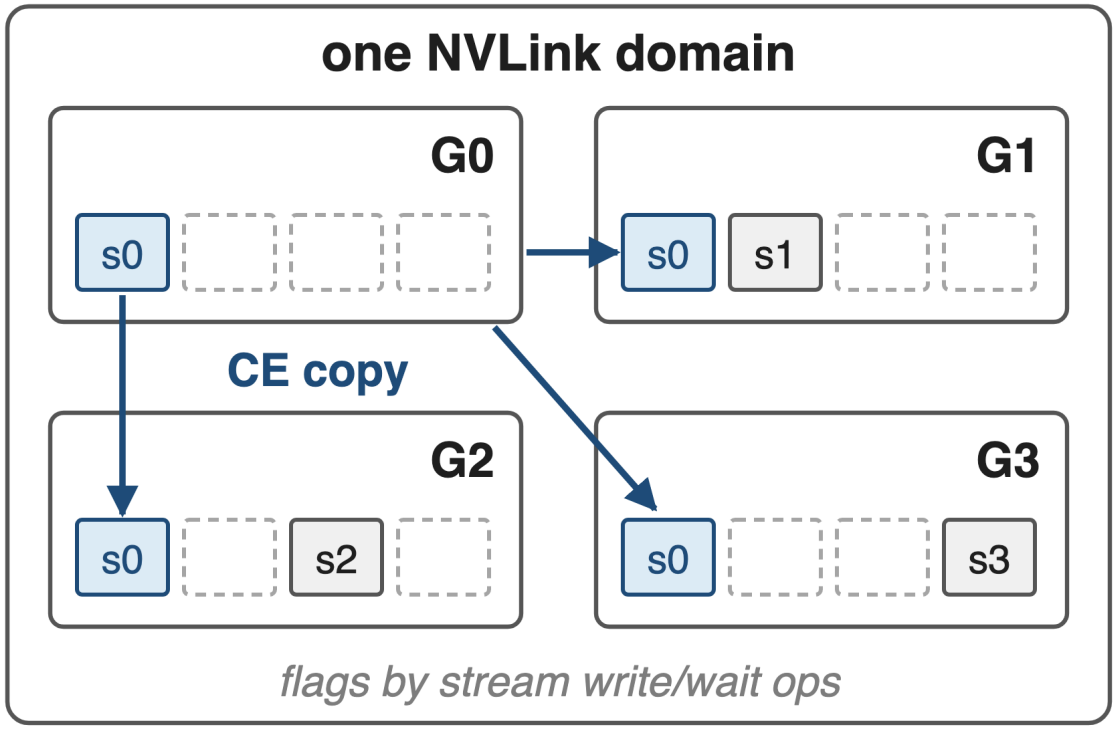


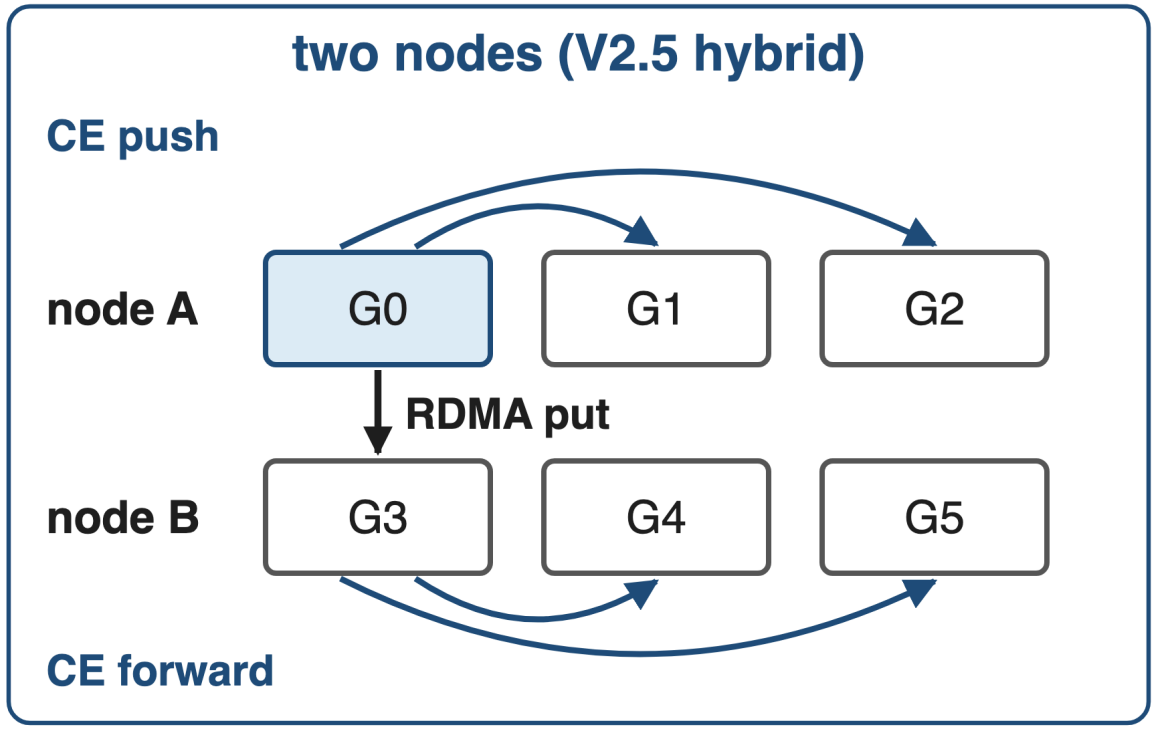
Kernels exit after posting; a slot ring and credits bound in-flight tensors

❷ CP all-gather runs on copy engines



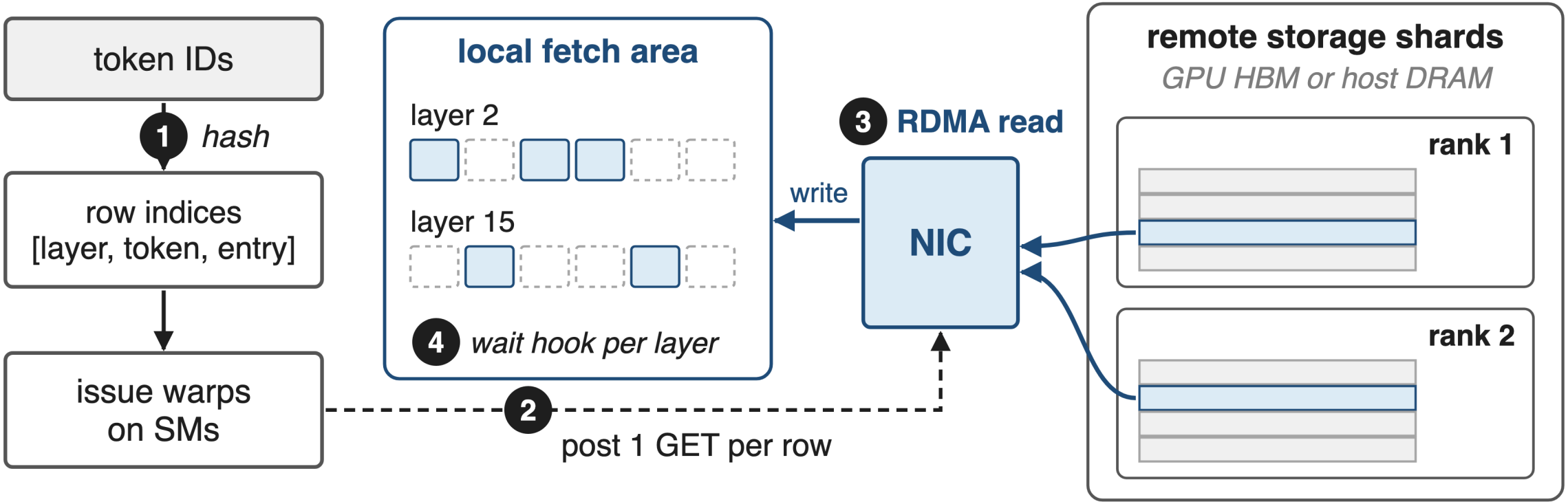
❷ CP all-gather runs on copy engines





NVLink path launches no kernel; across nodes CTAs only post RDMA puts

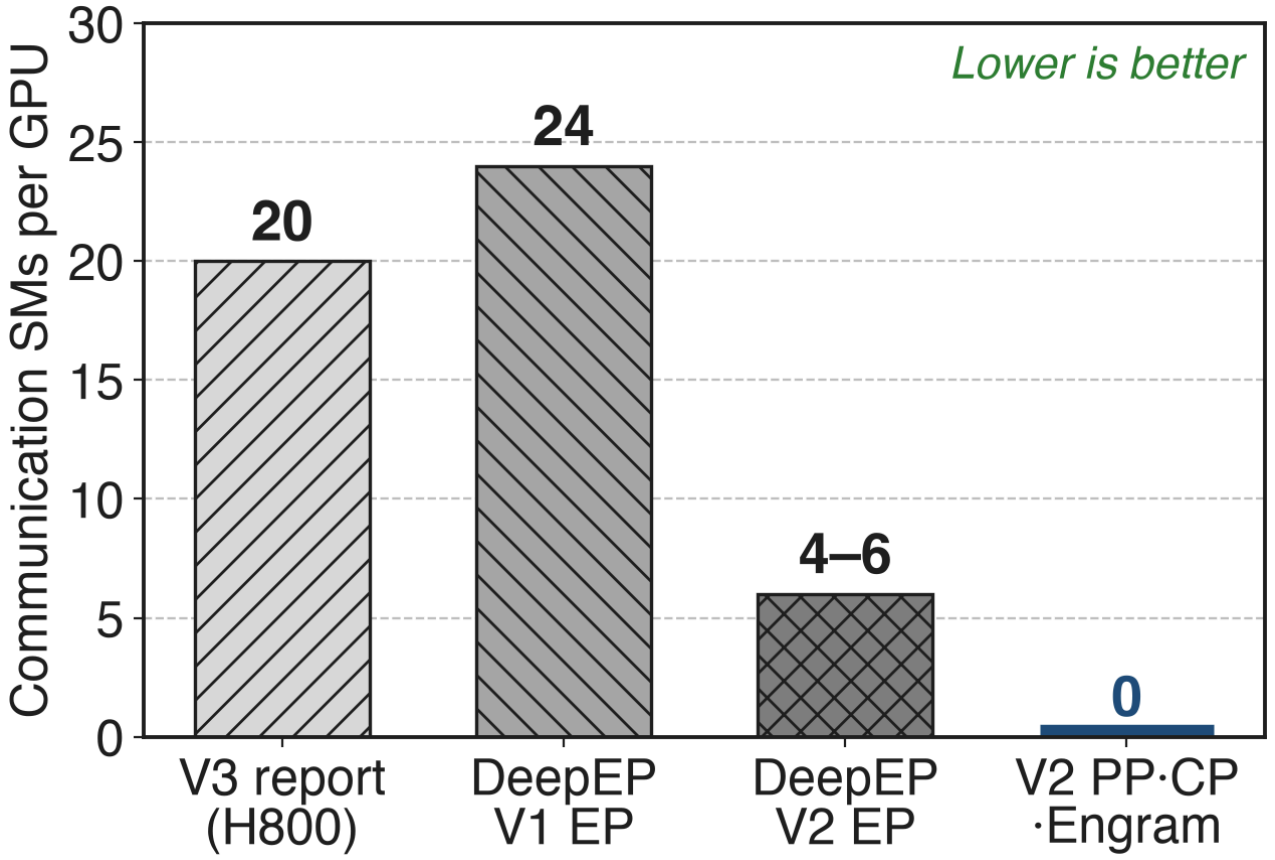
❸ Engram lookups become RDMA reads



Indices follow from token IDs, so fetches issue early and wait per layer

Transfers drop from 20 SMs to zero

- V2 EP: **4–6** SMs, down from **24**
- PP, CP, Engram: **0** SMs while bytes move
- No bandwidth numbers published yet



Only sums and routing still need SMs

payload moved by NIC or CE	
PP send / recv	RDMA put
Engram fetch	RDMA get
all-gather	CE or RDMA put

still needs SMs	
EP dispatch	routing + layout
EP combine	weighted sum
reduce-scatter	sum
all-reduce	sum

DeepEP V2.5: all-gather is copy-engine driven (num_sms=0); reduce-scatter, all-reduce, EP dispatch and combine use SMs

The 0-SM line is arithmetic: copies offload, sums stay on SMs

Zero SMs on the wire, not zero cost

- **Short kernels remain**: PP staging, Engram issue
- **Receiver spins** until the signal lands
- Extra copies through symmetric slots

- **Experimental**; V2.5 drops the "0 SM" label
- PP: RDMA-only group, adjacent ranks
- NCCL ≥ 2.32.3 with Gin; Hopper or newer

Summary

- DeepSeek-V3 gave communication **20 of 132** SMs
- **Insight**: copies need no SM; the NIC and copy engines move them
- PP put, CE all-gather, Engram get: **0** SMs on the wire (V2 EP: **4–6**)
- Experimental in DeepEP V2 and V2.5

https://github.com/deepseek-ai/DeepEP